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The day-of-the-week effect revisited: international evidence

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Abstract The aim of this study is to determine whether the DOW effect still exists, and to evaluate empirically the explanations of the DOW effect for international equity markets. Evaluating 51 markets in 33 countries for the period between January 2000 and December 2007, reveals that the DOW effect persists for a significant proportion of equity markets. Evaluating open-to-close returns, liquidity, size effect and possible spill-over effects, the DOW effect can be explained for almost of all the exchanges. Individual stock analysis, covering 37,631 stocks traded in 51 equity markets shows that *a DOW effect in returns exists for a statistically significant proportion of individual stocks in almost all of the markets in the study*. Even markets without a market-level DOW effect contain a surprisingly large proportion of stocks with individual-level DOW effects. Interestingly, this proportion is only marginally lower than that which is found in markets with a market-level DOW effect.

Keywords Day-of-the-Week Effect · Market Efficiency · Market Anomaly

JEL Classification G14 · G15

1 Introduction

The existence of consistent patterns in returns is an important marker of inefficiency and of arbitrage opportunities. Thus, calendar anomalies receive considerable attention from scholars as well as from practitioners. Earlier studies provided sufficient evidence of the day-of-the-week (DOW) effect in returns for U.S. equity and non-equity markets as well as for international markets. Recent studies, however, show that the DOW effect has disappeared in some of the developed markets and is disappearing in emerging markets.

There are three motivating questions for this study. *Are there still any market-level DOW effects? Are there DOW effects at the individual stock level? If so, what explains them?*

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The first question is: *are there any remaining market-level DOW effects?* On the market level, this study provides evidence that the DOW effect exists in 24 of 51 markets for Mondays and in 32 on Fridays.¹ It should be noted that there is a tendency in the literature to evaluate market wide averages for the DOW effect and/or indices comprised of a portfolio of stocks. Evaluation of individual stocks for a DOW effect is rare, which leads to the second question for this study.

Are there any DOW effects for individual stocks? At the individual stock level, the DOW effect exists for about 7% of stocks in each market. In fact, the difference between markets with and without DOW effects are quite minimal: (1) markets with a market-level DOW effect have, on average, 8.28% (8.37%) of their stocks with individual-level DOW effect on Mondays (Fridays), (2) markets without market-level DOW effects have, on average, approximately 7.25% (6.74%) of their stocks with individual-level DOW effects on Mondays (Fridays). *Thus, it is concluded that the DOW effect has not disappeared completely; instead, it has disappeared to the point that it cannot be detected using market averages.* A subsidiary question is: *if a market has no DOW effect in market returns, does that mean that a statistically insignificant number of stocks exhibit a DOW effect in those specific markets?* It could be argued that since the DOW effect has disappeared and some of the markets have no unexplained DOW effect left in them, there should either be no, or very few, stocks with a DOW effect. This is not the case. As we will show, some of the most important markets such as the NYSE show no DOW effect in market returns and yet approximately 9% of its stocks show a statistically significant² DOW effect in returns on Mondays and Fridays. Stocks with DOW effects tend to be smaller, and are therefore not detectable at the market-wide level. The overall evidence reveals that, on average, 7.73% (8.03%) of *all* stocks show DOW effect on Mondays (Fridays) at the 0.05 level. Moreover, as will be explained below, this result is not attributable to Type-I error.

The third question is: *what explains the DOW effect? Is it a true anomaly, or can we find its causes?* Hypotheses explaining the DOW effect include (among others) “differences in settlement and trading days”, “the non-trading problem”, “institutional trading”, “spill-over effects” and “inefficient econometric methodology”. There is also evidence to suggest that the DOW effect is in “calendar time” but not in “trading time”. Additionally, persistent anomalies in liquidity may be to blame for the DOW effect. Since equity markets around the world have diverse microstructures and clientele, there may be no one-size-fits-all explanation for the DOW effect. What is important, however, is that the DOW effect in market returns can be explained—as we will show—for almost all equity exchanges using one of the evaluated hypotheses here.

Based on data for 37,631 stocks traded in 51 equity markets in 33 countries³ for the period between January 2000–December 2007, this study contributes to the existing literature by providing evidence that the DOW effect in market returns

¹ While tables include results for Monday, Tuesdays, Thursdays and Fridays, for the purpose of compactness, discussion is centered especially on Mondays and Fridays. Whenever there are significant points to be made about Tuesdays and Thursdays then they are included in the discussion.

² For this study, statistical significance refers to statistical significance at 5% or better.

³ One of the challenges of the study is the reporting. The results provided with this study are a summary of the complete tests. All of the tables for market-wide and one-by-one tests are available upon request.

may seem to be disappearing but the DOW effect for a statistically significant number of individual stocks still exists. Furthermore, the hypotheses evaluated here fail to explain the reported DOW effect in individual stocks' returns.

While this study may be about DOW effect—perhaps one of the most reported market anomalies—our results apply to other market-level work. For instance, for studies that evaluate Granger type causality or cointegration between markets, the results could simply be due to a few select individual stocks which would present a completely different inference. The intent of the paper is not to limit the contribution to just DOW effect but to point out the potential differences between market level evaluations and individual stock level analyses. To investigate this, we employ a unique dataset consisting of bid-ask-spreads and returns of over 37,000 stocks. This allows us to evaluate almost every DOW hypothesis for even developing markets. This mass application, and its potential generality, are what make this study unique.

In the following section a brief summary of the previous literature is provided about the evidence of the DOW effect and related hypotheses. The third section provides the variable descriptions and the data. Definitions of hypotheses along with the econometric models are provided within the fourth section. Market level and individual stock level models are discussed with empirical results and their implications. An [Appendix](#) which includes the tables follows the concluding remarks of the study.

2 Previous literature

The DOW effect is defined as a pattern of relatively low returns for some days of the week (ex. Mondays) and relatively high returns for some other days of the week (ex. Fridays). Evidence for the existence of the DOW effect is mixed for different countries and different time periods. Fields (1931) presented some of the earliest evidence for the DOW effect. There have been numerous studies of the DOW effect in US markets⁴ and international markets.⁵ Each of these studies has either focused on markets in developed countries—such as those in the OECD—or has a regional focus. To our knowledge, no one has published a worldwide study, especially at the individual stock level.

There are different explanations for the occurrence of the DOW effect. While some of the hypotheses explain the DOW effect and argue that it is not an anomaly, others basically reject its existence. For instance, the *difference in settlement and trading days*, the *non-trading problem*, and the *institutional trading hypotheses* argue that the DOW effect exists but is not an anomaly. On the other hand, *inefficient econometric methodology*, *calendar time vs. trading time* and the *offsetting effect of liquidity* hypotheses argue that the DOW effect does not exist.

⁴ Some of the studies for U.S. markets include Fama (1965), Merrill (1966), Cross (1973), French (1980), Gibbons and Hess (1981), Keim and Stambaugh (1984), Lakonishok and Smidt (1988), Linn and Lockwood (1988) and Bessembinder and Hertzel (1993).

⁵ Studies of the DOW effect in international markets include Theobald and Price (1984), Jaffe and Westerfield (1985), Condoyanni et al. (1987), Aggarwal and Rivoli (1989), Chang et al. (1993), Tong (2000), Bayar and Kan (2002) and Ajayi et al. (2004).

The *calendar time hypothesis* posits that the return generating process is continuous so that returns should be related to calendar days. This suggests that investments in equity markets should receive higher returns on Mondays as a compounded return for the weekend (calendar, non-trading) days. In fact, the Monday returns should be about three times as much as other weekdays. However, most of the studies find Monday returns to be lower than other weekdays (and negative).⁶

The *settlement issue* refers to the difference of days between settlement and trade. Considering settlement periods of less than five business days, later days of the week provide advantages to investors compared to earlier days of the week. Lakonishok and Levi (1982) and Dyl and Martin (1985) provide a partial explanation for the DOW effect along these lines.

The *institutional trading issue* refers to the existence (or lack thereof) of institutional trading on certain weekdays. Lakonishok and Maberly (1990) show that unlike institutional investors, individual investors have the tendency to trade on Mondays. Their findings provide some explanation for the DOW effect. The institutional trading hypothesis posits that individual investors obtain and process information after trading hours, especially over the weekend. Increased individual investor activity on Monday leads to the Monday effect. The lack of favorable analyst reports over the weekend, with which to sustain individual investors' long positions, results in negative Monday returns. The preference of companies to leave the bad news to weekends also contributes to negative Monday returns.

The *calendar trading hypothesis* posits that the returns are based on calendar days and should be higher on Mondays to compensate for the weekend returns. The *Trading time hypothesis* posits that the returns are based on trading hours so there should be no differences between weekdays. A negative Monday effect using close-to-close returns would be contrary to the calendar time hypothesis. Also, a negative Monday effect using open-to-close would be contrary to the trading time hypothesis. In a study that evaluates trading and calendar time hypotheses, Rogalski (1984) decomposes Monday's returns (Friday Close to Monday close) into weekend return (Friday close to Monday open) and Monday trading return (Monday open to Monday close). Rogalski finds that the previously reported low (negative) Monday returns are actually weekend returns. Accordingly, there is no Monday effect when Monday trading returns are considered. The negative returns over the weekend are usually explained by the abundance of news released over the weekend. It is also argued that companies may prefer to delay bad news to weekends (non-trading holidays). In this paper, we will show that a DOW effect exists for 24 (32) markets for Mondays (Fridays). It is interesting to note that for 4 (7) of the markets, the DOW effect disappears for Mondays (Fridays) using open-to-close returns instead of close-to-close returns. More interestingly, for 9 (4) markets without DOW effect using close-to-close returns, a DOW effect exists using open-to-close returns; a clear rejection of the *trading time hypothesis* for these markets.

The *investor preference* and *clientele effect*, also referred to as the *non-trading problem*, use size as an explanatory factor of the DOW effect. Since there may be differences between trading small and large stocks, evaluating market indices that

⁶ Please refer to Pettengill (2003) for a detailed survey of the Monday effect. Some of the early evidence against the calendar day hypothesis was provided by Fama (1965), Merrill (1966) and Cross (1973)

combine these stocks may introduce a bias. Gibbons and Hess (1981), Keim and Stambaugh (1984) and Rogalski (1984) evaluate the size effect to test the non-trading problem. The market average used in estimating the DOW effect can either be equally weighted (EW) or value weighted (VW). Using an equal (value) weighted average, the DOW effect in smaller (larger) companies will have more impact on the overall market average. Thus, for each of the explanations evaluated, VW market average is used as well as EW market average. We find that switching from equally-weighted to value-weighted market averages reduces the number of markets shown to have a DOW effect. Specifically, 24 (32) markets have a Monday (Friday) effect when using EW; however this number drops to 15 (19) when using VW. This evidence implies that size is a significant factor explaining the DOW effect and it is more apparent for smaller stocks.

The *spill-over effect* refers to the existence of DOW effect in returns for a market primarily due to DOW effect in other markets' returns. There are two channels in which a spill-over effect can take place. First, if a market is Granger-caused by another market, it would be natural to expect that any anomaly in the "causing" market would be reflected as an anomaly in the "caused" market. Second, if two markets share similar international clientele, any anomaly due to behavioral aspects (i.e. trading preferences) of such clientele would appear on both markets. To test the spill-over effect, Granger causality is estimated for all of the equity markets included in the study to evaluate whether or not their returns are caused by others' returns. Causality from developed markets to developing markets would test the first channel of spill-over effect. Causality between markets with similar characteristics (developed with developed, developing with developing, within the same region, with similar risk levels and/or with similar market sizes etc.) would test the second channel⁷ of spill-over effect. The results are quite significant: all of the markets with a DOW effect in returns are caused by at least one of the markets in Japan, U.K. and/or U.S.

The *econometric methodology issue* refers to the reliability of estimations to test for the DOW effect. Connolly (1989) suggests that the methodology should adjust for large sample size and for heteroskedasticity. Estimating a Generalized Autoregressive Conditional Heteroskedasticity (GARCH⁸) model, he finds that sample size and heteroskedasticity provide some explanation for the DOW effect. Using a similar methodology Chang et al. (1993) found similar results in five different countries.

Based on the well documented relationship between returns and liquidity,⁹ it could be argued that the DOW effects in returns and in liquidity are related. Admati and Pfleiderer (1988) and Chowdhry and Nanda (1991) emphasize that while return anomalies get arbitrated away, liquidity anomalies make investors more cautious, which in turn makes the liquidity anomalies larger and more persistent. Since liquidity and returns are correlated, the DOW effect in returns may in fact be due to the persistent seasonality in liquidity.

⁷ For the purpose of compactness, results for the Granger causality from the markets in Japan, U.K. and U. S. to all other markets are reported. Causality results between all markets included in the study are available upon request.

⁸ For ARCH, see Engle (1982) and for GARCH, Bollerslev (1986)

⁹ See, for example, Amihud and Mendelson (1982), Karpoff (1987), Hiemstra and Jones (1994), Eleswarapu (1997), Jacoby et al. (2000), Domowitz et al. (2001), Lee (2005), and Bekaert et al. (2007)

The *impact of liquidity* on the DOW effect in returns is evaluated by estimating the corresponding DOW effect in the daily change in percentage spreads. Lakonishok and Maberly (1990) evaluate the DOW effect and find that individuals have the tendency to sell on Mondays more than any other week day. There is a correlation between returns and liquidity, especially for developing markets. If markets have lower liquidity on, say, Mondays (Foster and Viswanathan 1993), a DOW effect in returns would also be expected on Mondays. Accordingly, out of the 24 markets that had a Monday DOW effect on EW returns, 18 also had a Monday DOW effect on liquidity. Nineteen of the 32 markets that had a Friday effect also had it on liquidity as well. Liquidity therefore seems to be an important factor explaining any existing DOW effect in market returns.

One of the main contributions of this study is to evaluate various hypotheses for the DOW effect in market returns and for individual stock returns. While it is expected that the evaluated hypotheses (independently or in combination) explain any DOW effect in market returns, they fall short of providing an explanation for any DOW effect in individual stocks' returns. The extent and uniqueness of the data allow for evaluation of several different explanations for the DOW effect for 51 stock exchanges in 33 countries for an extensive time period. Market capitalization as well as bid-ask spreads make it possible to evaluate size and liquidity for not only developed markets but also for developing markets. Finally, the two main contributions of the study are that: (1) the DOW effect in market returns is explainable for almost all of the equity markets included in the study and (2) the DOW effect is still present for a number of individual stocks (on average about 7%) traded on *all* markets.

3 Data

In this study, two types of variables are used: variables for individual stocks (*i*) and variables for the overall markets (*c*). The daily return (*R*) is defined as the log difference of daily closing prices. In order to test the *institutional trading*, *calendar trading* and *trading time* hypotheses, open-to-close returns are used as well as close-to-close returns. While the previous liquidity literature employs several measures of liquidity including quoted spread, effective spread, proportional spread and proportional effective spread, these measures of liquidity are highly correlated with each other. For the purpose of compactness percentage spreads are used which has the least amount of skepticism for time-series normality in terms of skewness, kurtosis and variance. Percentage spread (*PRS*) is defined as the bid-ask spread as a percentage of the daily closing price. Daily change in the liquidity measure is also defined as the log difference of daily values. For the market variables, value and equal weighting are used. Following Chordia et al. (2001), a different market variable is calculated for each stock by excluding the stock from its own market variable.

The data are obtained from Reuters through the "QuoteCenter" application via Equis International.¹⁰ The period for the data is between January 2000 and December 2007. The data are filtered for the specifics of this study following previous day-of-the-week effect and liquidity literature. Accordingly, the sample includes ordinary

¹⁰ <http://www.equis.com/>

stocks with at least 40 annual observations. We also follow Chordia et al. (2000) filter for the outliers for the excessive spreads and prices. A window of five standard deviations is employed to eliminate remaining outliers. Finally, a minimum of 30 stocks is needed to include each equity markets. The final dataset includes 37,631 stocks traded on 51 stock exchanges in 33 countries.¹¹

4 Hypotheses, econometric models and empirical results

4.1 Market level analysis

The initial step in the analysis is to test for the existence of the DOW effect at the market level. Since the main argument of the study is the continued existence of the DOW effect at the individual stock level, market level analysis is included primarily for comparison with the previous literature. These tests establish that for many of the equity markets the DOW effect has disappeared as expected (and as reported in the existing literature), however for some of the markets it still exists. They also establish a comparison for the argument that while markets may not have a DOW effect at the market level, they all have a number of individual stocks that do have a DOW effect. Therefore, the first hypothesis (*H1*) is the existence of the DOW effect in equally weighted market returns. It is tested using a GJR-GARCH (1, 1) model which became quite common after Glosten et al. (1993):

$$MR_{c,t} = \alpha_{1,c} + \alpha_{2,c}MR_{c,t-1} + \sum_{j=1}^4 \beta_{R_c}^j D_{c,t}^j + \varepsilon_{R_{c,t}} \quad (1)$$

$$\sigma_t^2 = \gamma_0 + \gamma_1 \sigma_{t-1}^2 + \gamma_2 \varepsilon_{t-1}^2 + \gamma_3 \varepsilon_{t-1}^2 I_{t-1}$$

where $I_{t-1}=1$ if $\varepsilon_{t-1}>0$ and $I_{t-1}=0$ otherwise.

$$\varepsilon_t | \delta_{t-1} \sim N(0, \sigma_t^2)$$

$$j = \{1 = \text{Monday}, 2 = \text{Tuesday}, 3 = \text{Thursday}, 4 = \text{Friday}\}$$

$MR_{c,t}$ denotes the market-level returns, either EW or VW, for market c . D is daily dummy variable which has a value of one for the day it is assigned and zero otherwise. The close-to-close returns column in the Table 1, provides the coefficients for daily dummy variables for each of the equity markets included in the study. For 24 (32) of the markets, DOW effect is statistically significant² for Mondays¹ (Fridays). Sixteen of the 24 markets with DOW effect on Monday have negative coefficients. Thirty one of the 32 exchanges with Friday effect have positive coefficients. These results are in line with the expectations about the disappearance of the DOW effect in returns. Some of the most important equity markets such as the NYSE, Tokyo SE and London SE appear to be free from a Monday DOW effect. The NYSE and Tokyo SE also appear to be free from a Friday DOW effect. There is no apparent market characteristic or region to show DOW effect for either Monday or Friday. The tests

¹¹ Descriptive statistics are available upon request.

for *H1* have mixed results; the *DOW* effect has disappeared for some of the equity markets but still exists for some others.

Institutional trading, calendar trading and trading time arguments are tested as the second hypothesis (*H2*) by using open-to-close returns instead of the close-to-close returns for the Eq. (1). *H2* posits that the *DOW* effect should disappear when open-to-close returns are used since all three arguments are based on asymmetric information during and after trading. Open-to-close returns column in the Table 1, provides the coefficients for daily dummy variables for each of the equity markets. There are 29 markets with *DOW* effect on Monday and Friday using open-to-close returns. While for some of the markets *H2* is confirmed, for others it is clearly rejected. The results for *H2* reveal that the *DOW* effect is pronounced in two forms: (1) *DOW* effect using overnight (close-to-close) returns and (2) *DOW* effect using intraday (open-to-close) returns. These two forms of the *DOW* effect are expected to have different dynamics and therefore further hypotheses will be evaluated for both types of *DOW* effects. The tests for *H2* also have mixed results; *DOW* effects exist for close-to-close returns as well as open-to-close returns for some of the equity markets.

The *investor preference, clientele effect and non-trading problem* arguments are tested as the third hypothesis (*H3*) by using value weighted market returns instead of equally weighted market returns in Eq. (1). The results for the *H3* are quite interesting and establish the basis for the size effect. Table 2 provides the coefficients for the daily dummy variables for Eq. (1) with value weighted market returns. In terms of the close-to-close returns, the 24 (32) markets with *DOW* effect on Mondays (Fridays) using EW averages is reduced to 15 (19) with the use of VW averages. For the open-to-close returns, 29 markets with *DOW* effect on Mondays or Fridays with EW returns is reduced to 18 using VW returns. *With the use of VW market averages, the number of markets with DOW effect is significantly reduced which implies that DOW effect is more of an issue for smaller companies.*

The *impact of liquidity* hypothesis is tested by corresponding *DOW* effects in returns and in liquidity for the same weekday for the same equity markets. Thus, the fourth hypothesis (*H4*) is the existence of *DOW* effect in liquidity, measured on equally weighted and value weighted bases, and estimated by the GJR-GARCH(1,1) models below. (To conserve space, the equations for the conditional variance of the residuals are not provided in the model specifications below, as they are the same as in Eq. (1) above.)

$$ML_{c,t} = \alpha_{1,c} + \alpha_{2,c}ML_{c,t-1} + \sum_{j=1}^4 \beta_{L_c}^j D_{c,t}^j + \varepsilon_{L_c,t} \quad (2)$$

Coefficients for daily dummy variables for daily change in market-level liquidity (percentage spreads) using EW and VW market averages are provided in Table 3. Using EW market averages, 31 exchanges have a *DOW* effect in liquidity on Mondays, and 25 on Fridays. Also for the VW market averages, there are 27 exchanges have a *DOW* effect in liquidity on Mondays and 20 on Fridays. *The size effect seems to hold for the DOW effect in liquidity as well.* More importantly, in terms of the EW market averages, 18 (19) of the 24 (32) markets with *DOW* effect in close-to-close returns on Mondays (Fridays) have corresponding *DOW* effect in liquidity for the same day. For the open-to-close returns and EW market averages, 19 (15) of

the 29 (29) markets with DOW effect in close-to-close returns on Mondays (Fridays) have corresponding DOW effect in liquidity for the same day. The test results for *H4* imply that *DOW effect in returns and in liquidity are in fact corresponding for some of the markets but not for all*.

The *Spill-over effect* is examined using a VAR model. While the model is estimated between all countries, causation from the most influential markets based in Japan, U.K. and U.S. (to all other markets) are provided here. Therefore the fifth hypothesis (*H5*) posits that there is Granger causality from developed equity markets towards other equity markets. Two versions are tested, one using close-to-close returns, and a second using open-to-close returns:

$$MR_{c1,t} = \alpha_{1,c} + \sum_{i=1}^m \alpha_{2,i,c} MR_{c1,t-i} + \sum_{i=1}^m \alpha_{3,i,c} MR_{c2,t-i} + \varepsilon_{c1,t} \quad (3)$$

In Eq. (3) *c1* refers to the caused equity market and *c2* refers to the causing equity market (equity markets in Japan, U.K. and U.S.). The lag length, *m*, is calculated using a modified AIC criterion, and the maximum lag length is based on the Ng and Perron (2001) criterion. Granger causality is established by testing for the combined significance of the causing-markets' coefficients ($\alpha_{3,1,c} = \alpha_{3,2,c} = \dots = \alpha_{3,m,c}$) using a Wald test.

The results for the Wald test for the two versions of the Eq. (3) are provided in Tables 4 (close-to-close returns) and 5 (open-to-close returns). It is interesting to note that the causal relationship between the markets is more apparent using close-to-close returns compared to open-to-close returns. In either case, however, all of the markets that have a DOW effect in returns are Granger-caused in returns by at least one of the 11 markets (that are in Japan, U.K. and U.S.). These results *imply that the DOW effect may be the result of spill-over from some of the more developed markets*.

In sum, for the market-level DOW effect, it is evident that there is a DOW effect present in market returns for some of the equity markets. However, evaluation of the related hypotheses reveals that the DOW effect can be explained for almost all of the equity exchanges. The corresponding DOW effect in liquidity, size, spill-over effects and asymmetric information between trading and non-trading periods are the main explanations for the market level DOW effect in returns.

4.2 Individual stock level analysis

The evidence thus far shows that the DOW effect has disappeared for some markets, but still exists for others. The hypotheses evaluated here are quite successful in providing explanations for the existence of the DOW effect in market returns. In fact, for all but a few markets, the DOW effect is explained at the market level. However, since market averages were used, it has not—until this time—been possible to conclude that the DOW effect has disappeared completely. Since after reporting the anomalies the DOW effects seem to have disappeared, and there is a tendency in the existing literature to evaluate only market returns, any DOW effect in individual stocks' returns may have been overlooked and may still exist. The sixth hypothesis (*H6*) of the study is the existence of DOW effects in individual stocks' returns. The

following models are estimated for close-to-close and open-to-close returns of individual stocks listed on each of the equity markets:

$$R_{c,i,t} = \alpha_{1,c,i} + \alpha_{2,c,i}R_{c,i,t-1} + \alpha_{3,c,i}MR_{c,i,t} + \alpha_{4,c,i}\Delta(R_{c,i,t}^2) + \sum_{j=1}^4 \beta_{R_c}^j D_{c,t}^j + \varepsilon_{R_{c,i,t}} \quad (4)$$

With Eq. (4), the existence of the DOW effect is evaluated for close-to-close and open-to-close returns. In this equation, market risk is controlled with the contemporaneous VW market returns. Also, to control for return volatility, following Chordia et al. (2000), the daily change in squared return is included in the model.

The individual results for DOW effect in returns are impossible to report here, however, Table 6 provides a summary of these estimations. Accordingly, the number of stocks with statistically significant daily dummy variables is provided as a percentage of each of the equity markets. For instance, 7.68% of stocks in the Australian market, have a statistically significant dummy variable for Monday. Also, the cross-sectional mean for these statistically significant coefficients is provided under the β columns (for the Australian market, for Monday, the mean coefficient is -0.15%). *These results are the main aim of this study.*

In terms of the close-to-close returns, on average 7.73% of all stocks have statistically significant dummy variables for Mondays, and 8.03% for Fridays. For the open-to-close returns, on average 6.80% (6.35%) of all stocks have a statistically significant dummy variable for Mondays (Fridays).

More interestingly, on average for the markets that do not have DOW effect on Mondays (Fridays) using EW close-to-close returns, 7.25% (6.74%) of the stocks show DOW effects on Mondays (Fridays). Using open-to-close returns, 6.63% (5.39%) of their stocks show DOW effects on Mondays (Fridays). The implication for $H6$ is that a DOW effect in returns exists for individual stocks.

While a DOW effect for about 7% of the overall stocks may seem like a small percentage, the importance is emphasized through comparison. For instance, for the American SE, 7.24% of the market has a DOW effect in returns on Mondays. The American SE has a DOW effect in market returns on Mondays using EW, VW, close-to-close or open-to-close returns. A conclusion based on market returns would deem American SE as a market that shows DOW effect in market returns. On the other hand, the NYSE does not have a market-level DOW effect in returns on Mondays using EW, VW, close-to-close and open-to-close returns. However, 9.10% of stocks in the NYSE have Monday DOW effects.

Of course, when testing at the 5% level, as we do, Type-I error would be expected to lead to the spurious result that 5% of stocks in each market were found to have statistically significant DOW effects, even if no DOW effects were really present. The expected number of such “false positives” is 5%, but in finite samples this number might be slightly higher or lower. If our study found a market where 6% of stocks were found to have DOW effects, is this 6% proportion statistically greater than 5%? Are our results the result of Type-I error? Or, to phrase the question more precisely, do each of these markets exhibit a proportion of stocks with statistically significant DOW effects, that is statistically significantly greater than 5%? This sort of meta-level question requires that we turn to a Monte Carlo simulation in order to determine the sampling distribution of statistically significant daily DOW dummy variables.

For the Monte Carlo simulation, we use the average number of observations (1,150) and the average number of stocks (738) stocks along with the actual mean return and the actual standard deviation. The Monte Carlo experiment was estimated 10,000 times to generate a mean level and standard deviation of the percent of stocks with statistically significant weekday coefficients. Then, for each weekday and for each market, the percentage is tested for its statistical significance using the sampling distribution found through the Monte Carlo simulation. That is, the Monte Carlo means and standard deviations (for different weekdays) are used to test (using a *t*-test) whether each level for each weekday and each market is statistically significantly different than the level generated by the random data.

With a few exceptions, almost all percents for all markets and for all weekdays, are different than randomly generated levels at the 0.01 level. Thus, to rephrase the earlier question: are the results are attributable to Type-I error? No.

In order to evaluate and explain the DOW effect present in individual stocks, the seventh hypothesis (*H7*) of the study posits that the DOW effect in returns for individual stocks may be offset by a DOW effect in their own liquidity. Since on the market level, the DOW effect in liquidity was found to correspond with the DOW effect in returns, a similar hypothesis is tested for individual stocks. To test this hypothesis, the DOW effect in liquidity is estimated to correspond with the DOW effect in returns for the same weekdays for the same equity markets. The following equation was estimated, once for each of 37,631 stocks:

$$L_{c,i,t} = \alpha_{1,c,i} + \alpha_{2,c,i}L_{c,i,t-1} + \alpha_{3,c,i}ML_{c,i,t} + \sum_{j=1}^4 \beta_{L_c}^j D_{c,t}^j + \varepsilon_{L_{c,i,t}} \quad (5)$$

Obviously, we cannot report all of these results. Rather, we summarize them in Table 7. The figures in Table 7 are the percentage of each market with statistically significant daily DOW effect in returns and no corresponding DOW effect in liquidity for the same weekdays. For instance, for the Australian market, 6.92% of the market have statistically significant dummy variable for Mondays and do not have statistically significant daily dummy variables for liquidity for Mondays. On average the proportion of the market with statistically significant daily dummy variable for Monday (Friday) is reduced to 6.91% (7.23%) using close-to-close returns. Also for the open-to-close returns, level of statistically significant daily dummy variable for Monday (Friday) is reduced to 5.95% (5.67%). *The impact of the DOW effect on individual stocks' liquidity is small.*

5 Implications

Are there any remaining market-level DOW effects? The literature on the DOW has focused overwhelmingly on market-wide aggregates or sectoral indices. The DOW effect is one of many market anomalies that may lead to profitable trading rules. It would be expected that any existing DOW effects would have been arbitrated away. However, on the market level, this study provides evidence that the DOW effect exists in approximately half of the world's equity markets. The economic implications of this evidence are that not all equity markets are efficient, and profit opportunities still exist for investors even at the market level.

The DOW effect has not disappeared completely; instead, in many markets it has disappeared to the point that it cannot be detected using market averages. Are there any DOW effects for individual stocks? Evaluation of individual stocks for a DOW effect is rare, and represents one of the key contributions of this paper. The DOW effect exists for about 7% of individual stocks in each market (and even more in the NYSE). A Monte-Carlo simulation indicates that this result is not attributable to Type-I error. The implications of these findings are: 1) individual stocks provide profitable trading rule opportunities even for phenomena as widely reported as the DOW effect, and 2) researchers and regulators should be aware of the fact that individual stock level results could be completely different than results that are based on market level research (ex: 9% of the NYSE consists of potentially exploitable stocks while the NYSE as a whole shows no market-level DOW effect).

What explains the DOW effect? Is it a true anomaly, or can we find its causes? Several hypotheses explain the DOW effect, as discussed in previous sections. The DOW effect in market returns can be explained—as we have shown—for almost all equity exchanges using one of the evaluated hypotheses here. However, no one hypothesis is able to explain all reported DOW effect results. Interestingly, liquidity seems to be an important factor explaining any existing DOW effect in market returns. The implication is that there is a need for future research to explain anomalies that do not disappear even after reporting. Liquidity seems to be a variable of interest since liquidity anomalies are persistent and returns are related to liquidity. Tellingly, all of the markets that have a DOW effect in returns are Granger-caused by at least one of the equity markets in Japan, U.K. or the U.S. which are the most liquid markets in the world.

6 Concluding remarks

This study asks two questions: (1) *are there any market level DOW effects?* and (2) *are there any DOW effects for individual stocks?* When a market is deemed DOW effect free (or that the market-level DOW effect is explained), *is it really free of anomalous DOW effects when examined at the level of individual stocks?* Each market within the sample has a number of stocks with DOW effects in returns.

In the first part of the study, it was established that the DOW effect in market returns is explainable based on several factors: size, spill-over effect from other markets, concurrence with DOW effect in liquidity and asymmetric information between trading and non-trading periods.

In the second part, it was established that a DOW effect in individual stocks' returns still exists. In fact it exists for all of the equity markets included in the study. For the markets where market-level DOW effect was deemed non-existing, 7.25% (6.74%) of stocks show DOW effect on individual stock returns on Mondays (Fridays). This is compared to the level of the markets where market-level DOW effect still exists; 8.28% (8.37%) of stocks show DOW effect on individual stock returns on Mondays (Fridays).

It is common in the previous literature to evaluate composite market indices or indices of some group of stocks. Also, after reporting an anomaly it usually disappears. Thus, with the evidence presented here, it is concluded that the DOW effect in returns has disappeared to the point that studies that are based on market averages would not detect it. Also, the hypotheses that explain the market-level DOW effect do not explain the DOW effect in individual stocks' returns.

Appendix

Table 1 Coefficients for the day-of-the-week effect in equally weighted market returns estimated using Eq. (1) with close-to-close and open-to-close returns. Only the coefficients for the daily dummy variables are reported for compactness. “***”, “**”, and “*” refers to statistical significance at 1%, 5% and 10% level respectively. “+” indicates using GARCH(1,1) model instead of GJR-GARCH(1,1) due to non-convergence. Data were provided by Reuters through “QuoteCenter” application via Equis International for the period 2000–2007

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Australia, National A.T.	-0.07%***	-0.10%***	0.04%*	0.17%***	-0.15%***	-0.08%***	0.02%	0.12%***
Austria, Vienna SE	-0.02%	0.08%*	0.04%	0.10%*	-0.03%	0.01%	-0.02%	0.04%***
Belgium, Euronext Brussels	0.22%***	0.05%	0.24%***	0.25%***	-0.03%	-0.07%	0.01%	0.04%
Canada, TSX Venture E.	-0.49%***	-0.33%***	0.05%	0.40%***	-0.36%***	-0.14%***	0.02%	0.25%***
Canada, Toronto SE	-0.21%***	-0.17%***	0.04%	0.19%***	-0.25%***	-0.13%***	0.00%	0.18%***
Chile, Santiago SE	-0.06%***	-0.05%*	-0.01%	0.01%	-0.06%***	-0.04%***	-0.01%	-0.02%*
China, Shanghai SE	-0.03%	0.11%	-0.25%***	-0.14%	-0.04%	0.05%	-0.22%***	-0.15%
China, Shenzhen SE	-0.04%	0.12%	-0.26%***	-0.13%	-0.04%	0.06%	-0.24%***	-0.14%
Denmark, Copenhagen SE	-0.03%	-0.05%*	-0.03%	0.09%***	-0.03%*	-0.03%	-0.03%*	0.07%***
Egypt, Cairo SE	-0.11%*	-0.08%	0.06%	-0.21%	-0.11%***	-0.02%	0.04%	-0.05%
Finland, Helsinki SE	0.00%	-0.03%	0.03%	0.10%*	-0.05%*	-0.04%	-0.01%	0.06%***
France, Euronext Paris	0.00%	-0.04%	0.04%	0.11%*	0.02%	-0.02%	0.05%	0.07%***
Germany, Berlin SE	0.04%	0.01%	0.05%	0.04%	0.00%	0.00%	0.02%	0.01%
Germany, Stuttgart SE	0.10%	0.02%	0.02%	0.05%	0.03%	0.01%	0.05%	0.09%
Germany, Munich SE	0.12%***	0.16%***	0.18%***	0.14%***	-0.03%***	-0.02%	-0.01%	-0.01%
Germany, Frankfurt SE	0.08%***	-0.01%	0.05%	0.07%*	-0.02%	-0.02%	0.00%	0.03%*
Germany, XETRA	-0.09%	-0.13%	-0.05%	0.13%	-0.09%*	-0.08%*	-0.09%*	0.09%*
Germany, Duesseldorf SE	0.04%	0.03%	0.02%	0.01%	0.00%	-0.02%	0.03%	0.07%*

Table 1 (continued)

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Germany, Hamburg SE	0.03%	0.03%	0.06%	0.07%	-0.01%	0.06%	0.02%	0.05%
Greece, Athens SE	-0.32%***	-0.21%***	0.04%	0.10%	-0.25%***	-0.15%***	-0.09%	0.02%
Hong Kong, Hong Kong SE	0.03%	-0.06%	0.04%	0.14%***	0.00%	-0.05%	0.00%	0.12%***
India, Bombay SE	0.16%***	-0.22%***	-0.01%	-0.10%	-0.09%	-0.10%*	-0.04%	-0.02%
India, Indian N.E.	-0.05%	-0.34%***	-0.15%**	-0.14%**	0.01%	-0.04%	0.02%	0.02%
Indonesia, Jakarta SE	-0.32%***	-0.03%	0.00%	0.11%*	-0.18%***	0.00%	-0.03%	0.04%
Italy, Milano SE	0.00%	-0.08%**	-0.02%	0.00%	-0.05%*	-0.08%***	-0.05%*	-0.01%
Japan, Tokyo SE	0.02%	-0.08%*	0.00%	0.07%*	-0.06%*	-0.07%***	-0.01%	0.07%***
Japan, Osaka SE	0.11%***	-0.05%	0.03%	0.13%***	-0.10%***	-0.11%***	-0.05%*	0.05%*
Japan, Japan SDAQS	0.06%***	-0.06%**	0.01%	0.14%***	-0.06%***	-0.06%***	-0.02%	0.08%***
Japan, NASDAQ Japan	-0.29%***	-0.26%***	-0.10%	0.18%*	-0.31%***	-0.20%***	-0.11%	0.08%
Jordan, Amman F.M.	-0.14%***	-0.03%	-0.09%**	-0.05%	-0.12%***	0.06%**	0.01%	-0.17%***
Korea, Korea SE	-0.03%	-0.01%	0.16%**	0.10%	-0.01%	-0.08%	0.00%	0.08%
Korea, KOSDAQ	0.16%**	-0.06%	0.11%	0.14%*	0.20%***	-0.04%	0.02%	0.24%***
Malaysia, Kuala Lumpur SE	-0.30%***	-0.04%	-0.07%	0.10%**	-0.14%***	-0.03%	-0.04%	0.08%***
Netherlands, Euronext Amsterdam	0.13%***	0.05%	0.12%*	0.26%***	0.08%	0.02%	0.09%*	0.22%***
New Zealand, New Zealand SE	0.00%	-0.05%**	0.02%	0.05%*	-0.01%	-0.02%	0.00%	0.04%***
Norway, Oslo SE	-0.01%	-0.07%	0.06%	0.18%***	-0.02%	-0.04%	0.03%	0.12%***
Peru, Lima SE	-0.03%	-0.01%	0.01%	0.06%**	-0.03%**	-0.02%**	-0.01%	0.01%
Singapore, SE of Singapore	-0.24%***	0.01%	0.04%	0.16%**	-0.12%***	0.01%	-0.01%	0.13%***
South Africa, Johannesburg SE	0.03%	-0.02%	0.04%	0.09%***	0.00%	0.00%	0.05%***	0.07%***
Spain, Barcelona SE	0.03%	-0.06%	0.00%	0.05%	0.00%	-0.05%*	-0.02%	0.05%

Table 1 (continued)

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Sweden, Stockholm SE	-0.05%	-0.11%**	-0.01%	0.15%***	-0.03%	-0.05%	-0.01%	0.15%***
Switzerland, CHE SWX Swiss E.	-0.07%***	-0.03%	0.01%	0.09%***	-0.07%***	-0.04%***	-0.02%	0.06%***
Taiwan, Taiwan SE	-0.09%	-0.17%**	0.02%	0.14%**	-0.05%	-0.18%***	-0.05%	0.07%
Thailand, SE of Thailand	-0.22%***	-0.06%	-0.06%	0.05%	-0.14%***	-0.04%	-0.07%*	0.04%
U.K., SETS	-0.03%	-0.08%***	0.02%	0.10%***	-0.06%***	-0.07%***	0.01%	0.10%***
U.K., LSE-AIM	-0.02%	-0.11%***	0.05%***	0.12%***	-0.06%***	-0.09%***	0.06%***	0.11%***
U.K., IRS	-0.02% ⁺	-0.06% ⁺	0.01% ⁺	-0.02% ⁺	0.00%	-0.01%	0.00%	0.00%
U.S., New York SE	-0.06%	-0.01%	0.00%	-0.01%	-0.09%*	-0.02%	-0.03%	-0.06%
U.S., NASD OTCBB	-0.26%***	-0.11%***	0.01%	0.15%***	-0.23%***	-0.08%***	0.00%	0.10%***
U.S., American SE	-0.22%***	-0.07%**	0.01%	0.20%***	-0.20%***	-0.04%	0.01%	0.17%***
U.S., Nasdaq	-0.14%***	-0.07%	-0.02%	0.00%	-0.16%***	-0.07%	-0.04%	-0.02%

Table 2 Coefficients for the day-of-the-week effect in value weighted market returns estimated using Eq. (1) with close-to-close open-to-close returns. Only the coefficients for the daily dummy variables are reported for compactness. “***”, “**”, and “*” refers to statistical significance at 1% and 5% level respectively. “+” indicates using GARCH(1,1) model instead of GJR-GARCH(1,1) due to non-convergence. Data were provided by Reuters through “QuoteCenter” application via Equis International for the period 2000–2007

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Australia, National A.T.	0.08%**	0.02%	0.06%	0.09%**	0.02%	-0.01%	0.04%	0.07%**
Austria, Vienna SE	-0.13%+	0.05%+	-0.02%+	-0.07%+	-0.04%*	0.00%	-0.05%**	0.01%
Belgium, Euronext Brussels	0.21%**	0.08%	0.25%*	0.09%	0.01%	-0.01%	0.07%	0.00%
Canada, TSX Venture E.	-0.30%**	-0.38%**	0.15%**	0.49%**	-0.42%**	-0.22%**	0.10%**	0.38%**
Canada, Toronto SE	0.06%	-0.03%	0.02%	0.08%*	-0.02%	-0.06%	-0.04%	0.03%
Chile, Santiago SE	-0.19%**	-0.10%**	-0.03%	-0.04%	-0.18%**	-0.11%**	-0.05%	-0.09%**
China, Shanghai SE	-0.06%	0.08%	-0.22%**	-0.09%	-0.05%	0.05%	-0.20%**	-0.08%
China, Shenzhen SE	-0.05%	0.11%	-0.25%**	-0.09%	-0.04%	0.06%	-0.22%**	-0.10%
Denmark, Copenhagen SE	-0.02%	-0.06%	-0.02%	0.05%	0.01%	-0.05%	-0.04%	0.07%
Egypt, Cairo SE	-0.19%**	-0.14%	0.07%	-0.33%	-0.30%**	-0.11%	0.00%	-0.24%
Finland, Helsinki SE	0.04%	-0.04%	0.08%	0.01%	0.00%	-0.10%	-0.06%	-0.01%
France, Euronext Paris	0.11%	0.03%	0.11%	-0.01%	0.13%	-0.02%	0.15%	0.04%
Germany, Berlin SE	0.02%	0.02%	0.07%	0.02%	-0.02%	-0.01%	0.01%	0.02%
Germany, Stuttgart SE	0.04%	-0.33%	-0.19%	-0.07%	-0.14%*	-0.05%	-0.17%**	0.00%
Germany, Munich SE	0.09%*	0.20%**	0.15%**	0.12%*	-0.02%	-0.02%	-0.02%	-0.02%
Germany, Frankfurt SE	0.06%	-0.01%	0.06%	0.00%	-0.01%	-0.05%*	-0.02%	0.00%
Germany, XETRA	-0.18%	0.14%	-0.25%**	0.11%	0.06%**	0.02%	0.01%	0.02%
Germany, Duesseldorf SE	0.04%	0.09%	0.07%	-0.03%	-0.03%	-0.06%	0.00%	-0.02%
Germany, Hamburg SE	-0.01%	0.03%	0.05%	0.00%	-0.04%	-0.01%	-0.02%	-0.04%
Greece, Athens SE	-0.21%**	-0.14%**	0.06%	0.08%	-0.15%**	-0.11%**	-0.02%	0.04%

Table 2 (continued)

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Hong Kong, Hong Kong SE	0.00%	0.04%	0.05%	0.09%	0.03%	0.04%	-0.01%	0.05%
India, Bombay SE	0.05%	-0.09%	0.07%	0.02%	-0.10%	-0.02%	0.04%	0.02%
India, Indian N.E.	-0.03%	-0.02%	0.02%	-0.01%	-0.03%	-0.01%	0.06%	0.01%
Indonesia, Jakarta SE	-0.29%***	0.05%	0.00%	0.17%*	-0.13%*	-0.01%	-0.10%	0.16%**
Italy, Milano SE	-0.02%	-0.08%	-0.06%	-0.03%	-0.03%	-0.12%**	-0.08%*	-0.07%
Japan, Tokyo SE	0.00%	0.02%	0.04%	0.09%	-0.02%	-0.07%*	-0.01%	0.06%
Japan, Osaka SE	0.01%	0.01%	-0.01%	0.05%	0.01%	0.00%	0.03%	0.08%**
Japan, Japan SDAQS	-0.06%	-0.09%	0.05%	0.20%***	-0.16%***	-0.12%**	0.01%	0.13%***
Japan, NASDAQ Japan	-0.34%***	-0.17%	-0.04%	0.25%*	-0.28%***	-0.11%	-0.12%	0.18%
Jordan, Amman F.M.	-0.10%**	0.02%	-0.01%	-0.06%	-0.11%***	0.03%	-0.01%	-0.18%**
Korea, Korea SE	-0.05%	-0.01%	0.17%*	0.04%	0.01%	-0.08%	0.02%	0.02%
Korea, KOSDAQ	0.10%	-0.04%	0.17%*	0.12%	0.21%***	-0.09%	0.04%	0.29%***
Malaysia, Kuala Lumpur SE	-0.11%**	-0.02%	-0.02%	0.07%	-0.07%*	-0.02%	-0.03%	0.07%*
Netherlands, Euronext Amsterdam	0.14%	0.02%	0.26%*	0.32%**	0.09%	-0.01%	0.20%**	0.19%*
New Zealand, New Zealand SE	-0.02%	-0.01%	0.02%	0.01%	0.00%	-0.01%	-0.04%	-0.01%
Norway, Oslo SE	0.10%	-0.11%	0.16%*	0.22%***	0.03%	-0.15%**	0.08%	0.12%
Peru, Lima SE	0.04%	0.03%	0.01%	0.08%*	-0.04%*	0.02%	0.00%	0.02%
Singapore, SE of Singapore	-0.06%	0.02%	0.05%	0.13%**	-0.02%	0.01%	0.00%	0.10%*
South Africa, Johannesburg SE	0.20%***	0.01%	0.14%**	0.14%**	0.19%***	0.02%	0.11%**	0.12%**
Spain, Barcelona SE	0.04%	-0.07%	0.04%	0.06%	-0.04%	-0.09%*	-0.03%	-0.01%
Sweden, Stockholm SE	0.05%	-0.04%	0.04%	0.08%	0.05%	-0.08%	-0.03%	0.05%
Switzerland, CHE SWX Swiss E.	0.03%	-0.05%	0.04%	0.09%*	0.01%	-0.06%*	-0.03%	0.04%

Table 2 (continued)

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Taiwan, Taiwan SE	-0.11%	-0.11%	0.04%	0.09%	-0.05%	-0.07%	0.01%	0.09%
Thailand, SE of Thailand	-0.22%***	-0.07%	-0.11%	0.18%*	-0.17%**	-0.08%	-0.12%	0.13%
U.K., SETS	-0.02%	-0.08%**	0.01%	0.14%***	-0.04%	-0.08%**	-0.01%	0.13%***
U.K., LSE-AIM	0.00%	-0.07%*	0.16%***	0.21%***	-0.02%	-0.06%	0.15%***	0.21%***
U.K., IRS	-0.03%	-0.09%*	-0.01%	-0.04%	0.00%	-0.03%**	-0.01%	0.00%
U.S., New York SE	0.01%	-0.02%	0.00%	-0.02%	-0.03%	-0.04%	-0.04%	-0.10%**
U.S., NASD OTCBB	0.09%	0.14%	0.13%	0.29%***	0.08%	0.06%	-0.11%	0.53%***
U.S., American SE	-0.18%***	-0.03%	-0.12%*	0.18%***	-0.18%***	-0.03%	-0.13%***	0.11%*
U.S., Nasdaq	-0.15%*	-0.16%*	-0.13%*	-0.16%*	-0.14%*	-0.15%**	-0.10%	-0.20%***

Table 3 Coefficients for the day-of-the-week effect in equally weighted daily change in liquidity (percentage spread) estimated using Eq. (2) using equally-weighted and value-weighted market averages. Only the coefficients for the daily dummy variables are reported for compactness. “****”, “***”, “**”, and “*” refers to statistical significance at 1%, 5% and 10% level respectively. “***” indicates using GARCH(1,1) model instead of GJR-GARCH(1,1) due to non-convergence. Data were provided by Reuters through “QuoteCenter” application via Equis International for the period 2000–2007

Exchange	Equally weighted market average				Value weighted market average			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Australia, National A.T.	-2.11%****	-0.82%****	0.09%	1.51%****	-4.75%****	-1.73%	0.14%	0.98%
Austria, Vienna SE	0.60%	-1.21%	-0.11%	0.68%	-0.39%	-1.98%	0.24%	2.50%
Belgium, Euronext Brussels	-1.79%***	-1.62%**	0.29%	1.74%***	-5.04%***	-1.27%	0.25%	7.42%***
Canada, TSX Venture E.	2.13%****	0.29%	0.37%*	-0.03%	3.05%****	1.49%***	1.01%***	0.78%*
Canada, Toronto SE	0.83%****	-0.32%	0.69%***	1.92%****	-3.28%***	-5.33%****	0.00%	3.93%****
Chile, Santiago SE	-2.46%****	-0.39%	0.30%	0.25%	-5.85%****	-1.20%	-0.74%	1.56%
China, Shanghai SE	-0.61%	-0.40%	0.77%	0.49%	-0.06%	-0.36%	1.09%***	1.19%****
China, Shenzhen SE	-0.10%	-0.47%	0.48%	0.63%	-0.83%	-1.52%****	0.00%	0.18%
Denmark, Copenhagen SE	1.21%****	-0.35%	0.27%	0.52%	-2.87%***	-1.66%	0.14%	1.37%
Egypt, Cairo SE	-1.44%*	-1.20%*	-1.67%***	-1.23%	0.58%	-1.51%	-1.97%	-1.11%
Finland, Helsinki SE	-0.69%	-1.35%***	0.40%	0.37%	-3.73%*	-3.68%*	4.70%***	3.78%*
France, Euronext Paris	-1.64%****	-1.37%***	0.78%	0.87%	-6.39%****	-4.01%	-0.10%	3.98%
Germany, Berlin SE	0.13%*	-0.12%*	-0.01%	-0.05%	0.18%*	-0.22%***	0.01%	-0.08%
Germany, Stuttgart SE	-0.12%	-1.62%	-1.11%	-1.06%	3.97%	-3.75%	-4.35%***	-1.88%
Germany, Munich SE	0.31%	-0.11%	-0.26%***	-0.26%***	-0.05%	-0.27%*	0.07%	-0.14%
Germany, Frankfurt SE	-0.12%	-0.29%	0.51%***	0.61%***	-1.55%*	-0.76%	-0.11%	1.22%
Germany, XETRA	-1.51%	-2.72%*	-1.52%	-0.23%	-3.35%	-2.06%	-1.02%	2.89%
Germany, Duesseldorf SE	-0.34%*	-0.13%	-0.34%*	0.07%	-1.47%****	-0.25%	-0.25%	0.36%
Germany, Hamburg SE	-0.15%	-0.10%	0.21%	-0.31%	-0.06%	-0.04%	-0.10%	-0.21%***

Table 3 (continued)

Exchange	Equally weighted market average				Value weighted market average			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
Greece, Athens SE	0.92%*	-0.21%	0.58%	1.66%***	-3.99%***	-4.68%***	0.05%	1.28%
Hong Kong, Hong Kong SE	1.09%***	-0.48%*	-0.25%	-0.45%	0.47%	-0.18%	-0.53%	0.54%
India, Bombay SE	-1.36%***	-0.41%	-0.01%	1.91%***	-3.51%***	-2.11%	3.32%***	3.80%***
India, Indian N.E.	-0.23%	3.00%***	4.29%***	3.23%***	0.45%	5.35%***	7.21%***	6.50%***
Indonesia, Jakarta SE	3.57%***	-0.69%*	-0.22%	2.46%***	1.37%***	0.48%	0.32%	0.94%*
Italy, Milano SE	-0.49%	-0.62%	0.70%	0.94%	-3.09%***	-1.43%	0.32%	1.22%
Japan, Tokyo SE	-0.34%	-0.26%	0.13%	1.74%***	-1.95%*	-2.16%*	-1.19%	0.52%
Japan, Osaka SE	0.93%*	-0.72%	-0.05%	0.70%	-2.68%	0.25%	-1.93%	1.31%
Japan, Japan SDAQS	1.31%***	-0.20%	-0.09%	0.16%	-1.21%*	-1.32%***	-1.37%*	-1.96%***
Japan, NASDAQ Japan	-0.25%	1.05%	-0.16%	-1.13%	0.91%	0.52%	-0.06%	-1.62%
Jordan, Amman F.M.	-0.56%	-1.88%***	-0.68%	-0.67%	-0.80%	-3.09%***	-1.67%	-1.51%
Korea, Korea SE	-1.99%***	-0.44%	1.01%*	-1.10%*	-1.24%	-0.11%	3.30%***	-1.55%
Korea, KOSDAQ	-2.54%***	-0.82%*	-0.31%	-2.37%***	-3.70%***	-1.68%***	-0.43%	-2.96%***
Malaysia, Kuala Lumpur SE	4.29%***	-0.79%**	0.38%	0.88%***	1.20%***	0.31%	0.35%	0.69%*
Netherlands, Euronext Amsterdam	-1.37%	-0.56%	1.41%	1.57%*	-4.06%	0.52%	4.99%*	1.18%
New Zealand, New Zealand SE	-1.14%***	-0.64%	0.62%	1.60%***	-8.78%***	-3.94%***	-0.62%	-0.69%
Norway, Oslo SE	3.14%***	-1.08%***	0.60%	0.33%	-2.27%***	-2.58%***	-0.92%	0.05%
Peru, Lima SE	4.09%***	-0.52%	1.11%**	0.63%	3.87%*	-3.00%	-1.12%	-2.24%
Singapore, SE of Singapore	2.24%***	0.05%	0.13%	0.28%	-0.17%	0.37%	-0.23%	0.23%
South Africa, Johannesburg SE	-1.86%***	-2.03%***	1.58%***	5.52%***	-6.51%***	-2.45%	3.96%***	9.16%***
Spain, Barcelona SE	0.30%	-1.18%*	1.46%**	2.91%***	-0.33%	-0.75%	0.03%	2.22%*
Sweden, Stockholm SE	0.57%	-0.18%	0.87%**	1.29%***	-1.10%***	-0.83%	1.17%***	2.24%***

Table 3 (continued)

Exchange	Equally weighted market average					Value weighted market average				
	Monday	Tuesday	Thursday	Friday		Monday	Tuesday	Thursday	Friday	
Switzerland, CHE SWX Swiss E.	-1.75%*	-1.49%	0.54%	2.18%*		-2.98%**	-2.36%	-0.05%	0.86%	
Taiwan, Taiwan SE	1.09%***	0.73%*	0.47%	-0.19%		-0.03%	-0.11%	1.18%***	0.41%	
Thailand, SE of Thailand	3.32%***	0.72%**	0.78%**	0.47%		0.38%	0.26%	0.31%	-0.18%	
U.K., SETS	0.01%	-0.31%	0.32%	0.97%***		-1.72%**	-2.64%***	0.46%	2.64%***	
U.K., LSE-AIM	0.25%***	0.04%	-0.07%	-0.20%**		0.19%	-0.11%	0.23%	-0.51%*	
U.K., IRS	-0.54%***	-0.43%***	-0.36%***	0.16%*		-0.49%***	-0.52%***	0.37%***	0.36%***	
U.S., New York SE	-0.65%	-0.74%	0.58%	2.51%***		-1.30%	-1.53%*	0.31%	2.09%***	
U.S., NASD OTCBB	0.71%***	0.01%	0.06%	0.30%***		0.01%	-0.35%	-1.37%	-0.42%	
U.S., American SE	0.67%***	-0.28% ⁺	0.25% ⁺	0.90% ⁺ ***		-0.39%	-1.33%	0.51%	-0.11%	
U.S., Nasdaq	0.13%	-0.95%*	0.48%	0.78%		0.87%	-0.49%	1.04%	1.85%**	

Table 4 Wald test statistics for the Granger causality modes estimated with Eq. (3) using equally weighted close-to-close market returns. Markets in the first column are the caused markets whereas the markets in the other columns (Japan, U.K. and U.S.) are the causing markets. “***” and “**” refers to statistical significance at 1% and 5% level respectively. Data were provided by Reuters through “QuoteCenter” application via Equis International for the period 2000–2007

Caused markets	Japan			U.K.			U.S.				
	SDAQS	NASDAQ	Osaka SE	Tokyo SE	IRS	LSE- AIM	SETS	American SE	NASD OTCBB	Nasdaq	New York SE
Australia, National A.T.	12.99**	6.41	11.38***	12.64**	28.64***	58.75***	155.39***	582.99***	427.97***	613.69***	491.54***
Austria, Vienna SE	12.67**	5.10	19.22***	5.93	34.93***	50.66***	76.05***	81.49***	48.32***	67.49***	72.72***
Belgium, Euronext Brussels	2.33	1.34	6.03	0.42	2.93	1.80	3.45	6.01	7.60	6.45	13.47***
Canada, TSX Venture E.	6.51	3.59	5.30	5.61	10.55**	6.78	4.86	62.09***	32.60***	86.54***	63.06***
Canada, Toronto SE	14.45***	0.61	9.33	6.67	10.50**	13.64***	23.94***	31.35***	13.74***	94.40***	82.19***
Chile, Santiago SE	3.65	4.54	7.99	3.80	3.62	4.50	7.80	10.15**	8.68	11.92***	17.43***
China, Shanghai SE	3.41	3.22	3.40	3.15	7.53	2.73	8.64	6.74	2.55	6.15**	4.22
China, Shenzhen SE	3.25	2.47	4.04	3.53	6.84	3.70	10.60**	8.88	3.55	7.29**	4.65
Denmark, Copenhagen SE	7.11	6.34	9.40	3.64	8.32	1.92	5.93	172.15***	112.18***	314.76***	362.45***
Egypt, Cairo SE	7.62**	1.96	14.36***	5.30	10.69***	12.54***	12.02***	35.24***	17.45***	18.03***	16.20***
Finland, Helsinki SE	4.97	6.44	7.19	14.84***	12.14**	11.32**	5.19	214.16***	162.16***	455.07***	373.29***
France, Euronext Paris	3.90	5.60	1.58	3.19	7.17	1.06	2.96	151.00***	97.31***	272.50***	231.39***
Germany, Berlin SE	3.14	7.42	2.81	2.42	7.91	16.82***	38.86***	310.24***	143.78***	598.89***	593.60***
Germany, Stuttgart SE	8.53**	10.81**	13.03**	14.69***	47.67***	62.56***	58.72***	64.20***	52.41***	50.41***	46.46***
Germany, Munich SE	6.29	0.27	9.74**	13.02**	6.70	13.73***	19.23***	351.81***	249.08***	1211.73***	776.21***
Germany, Frankfurt SE	1.68	8.67**	0.72	3.16	20.49***	15.17***	39.75***	396.17***	278.71***	837.69***	569.18***
Germany, XETRA	6.87	20.10***	7.81	5.45	17.53***	31.67***	17.27***	58.44***	90.32***	78.31***	38.92***
Germany, Duesseldorf SE	9.53**	1.33	1.96	15.08***	10.82**	5.71	2.79	110.56***	93.44***	396.43***	243.04***
Germany, Hamburg SE	3.20	2.86	8.13**	2.19	40.51***	54.13***	115.14***	224.43***	138.08***	377.72***	348.45***
Greece, Athens SE	13.86***	16.94***	22.68***	7.62	26.35***	16.75***	27.94***	18.05***	18.26***	49.03***	62.63***

Table 4 (continued)

Caused markets	Japan			U.K.			U.S.				
	SDAQS	NASDAQ	Osaka SE	Tokyo SE	IRS	LSE- AIM	SETS	American SE	NASD OTCBB	Nasdaq	New York SE
Hong Kong, Hong Kong SE	23.43***	2.71	21.86***	15.53***	14.95***	44.87***	42.49***	201.57***	146.14***	299.51***	245.70***
India, Bombay SE	10.94**	11.91**	7.28	7.29	7.60	11.63**	9.72**	28.87***	21.71***	43.96***	40.14***
India, Indian N.E.	13.55***	22.56***	13.03***	9.57**	12.41**	23.79***	22.81***	43.05***	19.55***	63.28***	52.84***
Indonesia, Jakarta SE	16.38***	11.22**	19.72***	13.62***	3.35	32.77***	33.96***	45.52***	20.16***	85.56***	96.96***
Italy, Milano SE	7.30	0.39	4.70	9.53**	23.84***	5.95	12.18**	114.80***	72.79***	233.59***	252.05***
Jordan, Amman F.M.	9.61**	4.72	4.00	13.23***	12.36***	11.98**	15.22***	5.16	10.30**	6.82	6.45
Korea, Korea SE	6.26	4.52	8.73**	9.75***	40.92***	15.44**	28.36***	110.72***	63.97***	211.19***	198.64***
Korea, KOSDAQ	10.42**	1.34	14.07***	14.20***	32.92***	14.99***	25.18***	97.95***	53.01***	182.00***	159.93***
Malaysia, Kuala Lumpur SE	12.01***	1.00	26.40***	15.93***	19.24***	42.83***	38.33***	74.52***	74.18***	121.25***	98.78***
Netherlands, Euronext Amsterdam	4.30	2.90	2.91	0.76	15.11***	8.47	5.96	103.11***	62.66***	227.67***	241.42***
New Zealand, New Zealand SE	8.81	6.21	7.46	4.65	53.60***	47.01***	144.97***	286.25***	223.10***	441.73***	392.51***
Norway, Oslo SE	6.98	1.95	5.18	6.39	10.61**	4.35	7.16	217.32***	111.49***	302.21***	310.29***
Peru, Lima SE	8.22	1.66	1.76	3.78	5.02	2.26	4.56	7.88	9.20	14.10***	18.17***
Singapore, SE of Singapore	13.90***	4.56	18.54***	7.26	21.38***	34.18***	46.97***	101.20***	63.82***	147.01***	151.62***
South Africa, Johannesburg SE	8.19**	0.13	5.99	11.30**	2.66	1.62	15.56***	216.86***	136.74***	290.08***	288.70***
Spain, Barcelona SE	12.39**	7.25	7.07	6.98	5.19	5.89	5.78	145.26***	74.81***	217.35***	262.90***
Sweden, Stockholm SE	5.86	5.77	1.86	13.54***	4.40	8.98	4.75	124.80***	102.36***	326.78***	272.67***
Switzerland, CHE SWX Swiss E.	9.67**	3.73	7.08	4.95	11.66**	8.71	5.41	187.02***	90.23***	434.64***	520.54***
Taiwan, Taiwan SE	6.68	4.53	5.57	9.18	8.21**	7.23	14.01***	51.90***	37.64***	106.50***	103.48***
Thailand, SE of Thailand	6.40	8.57**	13.53***	0.19	5.18	22.80***	32.88***	61.60***	23.59***	63.17***	73.99***

Table 5 Wald test statistics for the Granger causality modes estimated with Eq. (3) using equally weighted open-to-close market returns. Markets in the first column are the caused markets whereas the markets in the other columns (Japan, U.K. and U.S.) are the causing markets. “***” and “**” refers to statistical significance at 1% and 5% level respectively. Data were provided by Reuters through “QuoteCenter” application via Equis International for the period 2000–2007

Caused markets	Japan			U.K.			U.S.				
	SDAQS	NASDAQ	Osaka SE	Tokyo SE	IRS	LSE- AIM	SETS	American SE	NASD OTCBB	Nasdaq	New York SE
Australia, National A.T.	10.16 **	10.24**	12.81**	6.29	6.21	46.27***	53.90***	91.72***	70.88***	97.99***	62.20***
Austria, Vienna SE	5.01	3.34	2.31	1.22	2.35	6.04	5.72	14.66***	14.25***	29.01***	31.71***
Belgium, Euronext Brussels	17.86***	7.39	13.76***	4.68	0.74	6.16	4.17	6.21	3.55	3.68	5.48
Canada, TSX Venture E.	4.27	1.29	5.34	3.69	16.38***	9.79**	5.62	42.33***	7.50	73.35***	50.13***
Canada, Toronto SE	11.15**	1.27	8.19	6.66	12.35**	8.47	24.87***	8.31	4.14	54.49***	32.82***
Chile, Santiago SE	4.62	4.41	6.95	3.73	1.77	2.50	2.90	4.81	1.86	3.88	4.46
China, Shanghai SE	1.92	2.95	1.44	2.21	0.47	3.56	5.14	0.56	1.23	0.90	0.72
China, Shenzhen SE	1.83	2.77	1.35	1.80	0.63	4.24	5.60	0.97	2.00	1.46	0.83
Denmark, Copenhagen SE	7.55	5.67	4.68	7.70	0.49	22.82***	11.38**	26.81***	26.20***	59.38***	56.23***
Egypt, Cairo SE	2.71	0.28	0.94	0.17	2.49	8.26	15.57***	27.88***	7.13	7.05**	12.67***
Finland, Helsinki SE	1.46	0.85	6.18	1.35	6.93	13.15**	0.82	50.61***	36.07***	105.34***	99.22***
France, Euronext Paris	0.26	0.65	0.01	0.33	1.26	4.78	2.00	29.45***	14.48***	35.59***	38.94***
Germany, Berlin SE	3.14	1.11	3.17	2.81	2.53	2.98	6.58	28.46***	16.17***	59.08***	52.24***
Germany, Stuttgart SE	4.69	6.22	6.47	6.41	0.38	4.39	9.82**	1.94	7.55	10.24**	10.30**
Germany, Munich SE	8.03	2.87	6.41	2.92	1.19	10.89**	26.21***	32.41***	16.44***	113.96***	69.68***
Germany, Frankfurt SE	4.25	1.75	4.34	3.25	2.86	18.88***	18.83***	37.15***	19.80***	81.91***	62.19***
Germany, XETRA	0.25	3.29	2.47	3.63	5.13	18.41***	10.34**	5.11	31.78***	17.85***	2.29
Germany, Duesseldorf SE	13.74***	2.62	7.67	4.05	5.38	36.98***	39.39***	35.50***	13.54***	91.59***	67.85***
Germany, Hamburg SE	3.53	0.93	5.34	4.19	1.73	4.93	2.03	8.48	3.08	5.30	4.57

Table 5 (continued)

Caused markets	Japan			U.K.			U.S.				
	Japan			U.K.			U.S.				
	SDAQS	NASDAQ	Osaka SE	Tokyo SE	IRS	LSE- AIM	SETS	American SE	NASD OTCBB	Nasdaq	New York SE
Greece, Athens SE	11.06**	11.38**	9.66**	7.16	22.23***	36.14***	57.36***	10.98**	28.43***	8.54	5.26
Hong Kong, Hong Kong SE	15.19***	5.55	10.62***	6.14	10.33**	57.45***	34.72***	29.90 ***	31.81 ***	40.19***	25.63***
India, Bombay SE	10.15 **	10.21 **	6.94 **	9.94 **	14.45***	19.41***	20.38***	27.41***	4.53	28.72***	34.62***
India, Indian N.E.	12.47**	9.75***	3.56	2.36	23.84***	27.39***	34.82***	36.72***	18.62***	41.11***	36.30***
Indonesia, Jakarta SE	5.92	3.44	13.62***	5.72	5.93	13.08**	25.19***	30.06**	26.61***	39.74***	35.01 ***
Italy, Milano SE	3.69	2.97	4.37	3.28	8.93**	2.71	8.07	10.65**	9.39	33.33***	26.28***
Jordan, Amman F.M.	4.40	9.25	4.24	3.42	10.59**	12.93**	2.75	10.43**	1.79	3.22	1.40
Korea, Korea SE	4.41	2.40	16.61	17.38***	7.03**	4.73	13.09**	7.31	11.64**	10.74**	8.79**
Korea, KOSDAQ	7.70	0.03	14.92***	24.38***	2.82	19.45***	23.64***	4.10	9.00	23.11***	27.80***
Malaysia, Kuala Lumpur SE	9.73**	5.08	10.62**	11.15**	3.92	31.42***	30.35***	13.81***	24.83***	22.56***	17.37***
Netherlands, Euronext Amsterdam	1.73	2.10	2.18	0.21	0.15	9.20	8.25	17.49***	5.90	30.06***	27.56***
New Zealand, New Zealand SE	5.41	5.16	4.53	2.33	7.33**	14.59***	45.53***	76.58***	34.74***	121.12***	123.87***
Norway, Oslo SE	5.75	5.55	4.43	1.45	2.53	15.13***	25.24***	16.02***	19.15***	25.98***	22.10***
Peru, Lima SE	2.80	3.35	1.71	1.98	5.15	1.37	3.72	5.50	6.10	2.08	2.86
Singapore, SE of Singapore	9.64**	8.09	14.78***	11.28**	5.36	31.41***	53.95***	26.03***	30.26***	59.93***	52.81 ***
South Africa, Johannesburg SE	13.22***	5.53	10.09***	2.61	3.75	4.20	7.53	45.75***	17.76***	57.00***	55.76***
Spain, Barcelona SE	10.78**	2.98	2.23	0.87	0.94	7.05	5.05	24.83***	20.11***	41.32***	39.54***
Sweden, Stockholm SE	1.26	3.02	2.25	5.77	12.81**	11.27**	4.02	12.25**	20.50***	39.45***	31.75***
Switzerland, CHE SWX Swiss E.	1.15	0.41	1.04	0.47	1.30	4.71	6.09	39.47***	30.24***	104.03***	89.36***
Taiwan, Taiwan SE	0.15	2.46	0.09	2.30	19.53***	4.23	23.81***	39.66***	28.24***	85.99***	101.60***
Thailand, SE of Thailand	1.31	0.77	0.44	0.24	3.54	31.48***	50.27***	27.97***	14.00***	32.05***	32.13***

Table 6 Results for the day-of-the-week effect in individual stocks' returns using close-to-close and open-to-close returns. β refers to the mean statistically significant coefficient (at 5% level) for the daily dummy and % refers to the percentage of the equity market's individual stocks with statistically significant daily dummy variables. For instance, 7.68% of National AT of Australia has statistically significant dummy variable for Monday and these coefficients have a mean of -0.15% . *** indicates that the percentage of individual stocks (that were found to have statistically significant DOW effects) was statistically significantly greater (at the 0.01 level) than what would be expected from Type-I error. Similarly, ** refers to significance at the 0.05 level, and * to 0.10 level. Data were provided by Reuters through "QuoteCenter" application via Equis International for the period 2000–2007

Exchange	Close-to-close returns						Open-to-close returns					
	Monday		Tuesday		Thursday		Friday		Monday		Tuesday	
	β	%	β	%	β	%	β	%	β	%	β	%
Australia, National A.T.	-0.15%	7.68%	-0.83%	6.27%	0.10%	5.47%	0.85%	9.83%	-0.36%	6.82%	-0.41%	6.72%
Austria, Vienna SE	0.04%	4.21%	1.12%	3.16%	0.93%	2.11%	0.11%	4.21%	-0.30%	3.16%		-0.49%
Belgium, Euronext Brussels	0.07%	9.28%	-0.30%	4.12%	-0.54%	4.64%	0.21%	9.79%	-0.12%	5.67%	-0.15%	3.09%
Canada, TSX Venture E.	0.25%	6.53%	-0.73%	5.38%	0.35%	5.33%	1.16%	9.30%	-0.62%	5.73%	-0.38%	6.53%
Canada, Toronto SE	-0.52%	6.90%	-0.77%	6.62%	0.51%	6.06%	0.96%	16.42%	-0.76%	11.29%	-0.55%	7.00%
Chile, Santiago SE	0.06%	6.20%	0.06%	4.65%	0.26%	3.10%	0.18%	6.20%	-0.18%	3.10%	-0.25%	3.10%
China, Shanghai SE	-0.14%	8.11%	0.18%	5.71%	-0.12%	4.79%	-0.28%	8.22%	-0.12%	8.33%	-0.03%	5.25%
China, Shenzhen SE	-0.27%	6.79%	-0.18%	8.60%	-0.43%	5.28%	-0.08%	6.94%	-0.33%	8.30%	-0.27%	7.54%
Denmark, Copenhagen SE	-0.13%	9.09%	0.03%	6.25%	-0.13%	6.82%	0.33%	9.66%	-0.39%	9.09%	0.17%	7.95%
Egypt, Cairo SE	-0.14%	7.81%	-0.77%	4.69%	-0.83%	5.47%			0.09%	3.91%	-0.02%	3.91%
Finland, Helsinki SE	0.28%	8.20%	0.16%	5.74%	-0.02%	5.74%	0.31%	13.11%	-0.38%	0.82%	-0.05%	5.74%
France, Euronext Paris	0.06%	5.44%	0.02%	5.82%	0.31%	6.32%	0.60%	8.34%	0.01%	4.93%	0.07%	5.31%
Germany, Berlin SE	0.32%	7.53%	0.16%	4.67%	0.10%	5.47%	0.27%	6.21%	-0.14%	5.07%	-0.04%	5.50%
Germany, Stuttgart SE	2.83%	3.35%	1.79%	3.77%	2.85%	3.35%	2.68%	4.18%	-0.52%	0.84%	-0.39%	2.93%
Germany, Munich SE	0.34%	6.88%	0.03%	6.49%	-0.07%	5.36%	0.37%	5.81%	-0.12%	4.91%	0.25%	6.09%
Germany, Frankfurt SE	-0.20%	6.17%	0.13%	5.74%	-0.63%	4.89%	0.37%	4.89%	0.23%	11.91%	0.40%	7.02%
Germany, XETRA	0.60%	4.08%	-2.48%	6.12%	-1.18%	10.20%	0.88%	10.20%	-0.27%	4.08%	-0.03%	6.12%

Table 7 Results for the day-of-the-week effect in individual stocks' returns using close-to-close and open-to-close returns for the stocks that do not have DOW effect in liquidity on the same weekday. β refers to the mean statistically significant coefficient(at 5% or lower level) for the daily dummy and% refers to the percentage of the equity market with statistically significant daily dummy variable. For instance, 6.92% of National AT of Australia has statistically significant dummy variable for Monday and these coefficients have a mean of -0.15% . Data were provided by Reuters through "QuoteCenter" application via Equis International for the period 2000–2007

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
	%	%	%	%	%	%	%	%
Australia, National A.T.	6.92%	5.77%	4.77%	8.98%	5.87%	6.02%	4.26%	8.03%
Austria, Vienna SE	4.21%	3.16%	2.11%	3.16%	2.11%	0.00%	2.11%	0.00%
Belgium, Euronext Brussels	8.25%	3.61%	4.64%	8.76%	4.12%	2.85%	6.19%	4.20%
Canada, TSX Venture E.	6.08%	5.03%	5.08%	8.34%	5.13%	6.10%	4.32%	5.85%
Canada, Toronto SE	6.34%	6.44%	5.69%	14.65%	10.73%	7.00%	4.94%	12.41%
Chile, Santiago SE	5.43%	4.65%	2.33%	5.43%	3.10%	2.90%	6.20%	3.28%
China, Shanghai SE	7.31%	5.25%	4.57%	7.19%	7.65%	5.15%	3.77%	7.80%
China, Shenzhen SE	6.18%	8.14%	4.83%	6.18%	7.99%	7.00%	2.87%	7.99%
Denmark, Copenhagen SE	7.39%	5.68%	6.82%	9.09%	8.52%	6.36%	5.11%	7.16%
Egypt, Cairo SE	7.81%	3.91%	4.69%		2.34%	3.35%	5.47%	
Finland, Helsinki SE	8.20%	5.74%	5.74%	13.11%	0.82%	5.19%	4.10%	7.49%
France, Euronext Paris	5.06%	4.93%	6.19%	7.33%	4.68%	4.84%	3.67%	3.67%
Germany, Berlin SE	5.99%	4.09%	4.58%	5.35%	4.80%	5.50%	3.63%	4.98%
Germany, Stuttgart SE	2.93%	3.77%	2.93%	2.93%	0.84%	2.58%	5.44%	2.44%
Germany, Munich SE	5.81%	5.81%	4.85%	5.19%	4.35%	5.91%	5.14%	5.04%
Germany, Frankfurt SE	4.89%	5.32%	4.04%	4.26%	9.79%	7.02%	5.11%	3.06%
Germany, XETRA	4.08%	6.12%	10.20%	8.16%	4.08%	5.08%	4.08%	
Germany, Duesseldorf SE	4.93%	4.93%	4.68%	6.07%	5.69%	4.66%	5.31%	6.78%
Germany, Hamburg SE	9.38%	3.13%	3.13%	7.29%	5.21%	9.26%	6.25%	4.88%
Greece, Athens SE	9.76%	4.57%	3.66%	8.54%	9.15%	2.62%	3.35%	4.86%
Hong Kong, Hong Kong SE	7.18%	4.44%	4.44%	4.87%	5.98%	5.17%	5.04%	4.43%
India, Bombay SE	9.30%	6.71%	4.93%	5.01%	6.87%	6.79%	5.01%	3.88%
India, Indian N.E.	9.18%	17.05%	13.11%	10.82%	4.26%	8.02%	7.21%	4.63%
Indonesia, Jakarta SE	5.00%	5.31%	3.44%	2.50%	4.69%	5.06%	6.56%	4.38%
Italy, Milano SE	8.21%	6.79%	4.29%	8.57%	5.36%	7.25%	5.00%	7.38%
Japan, Tokyo SE	6.08%	5.65%	4.67%	4.85%	7.56%	5.29%	5.10%	4.76%
Japan, Osaka SE	8.62%	4.41%	5.61%	5.41%	6.41%	6.09%	4.81%	5.21%
Japan, Japan SDAQS	8.27%	4.83%	4.51%	5.26%	5.48%	5.37%	5.69%	4.29%
Japan, NASDAQ Japan	10.20%	5.44%	6.12%	8.84%	8.16%	6.73%	5.44%	3.50%
Jordan, Amman F.M.	6.21%	5.59%	5.59%		8.07%	5.83%	2.48%	
Korea, Korea SE	5.49%	3.98%	3.98%	6.56%	7.00%	4.16%	5.90%	6.66%
Korea, KOSDAQ	5.44%	4.93%	4.93%	3.82%	7.80%	4.85%	4.41%	4.12%
Malaysia, Kuala Lumpur SE	9.58%	7.04%	6.10%	6.30%	6.85%	5.26%	4.51%	6.38%
Netherlands, Euronext Amsterdam	7.63%	2.29%	6.11%	11.98%	3.82%	2.04%	6.87%	11.45%
New Zealand, New Zealand SE	7.69%	6.15%	4.10%	6.48%	3.59%	4.62%	3.59%	5.47%
Norway, Oslo SE	5.04%	5.88%	9.66%	13.07%	5.46%	5.88%	5.88%	8.12%

Table 7 (continued)

Exchange	Close-to-close returns				Open-to-close returns			
	Monday	Tuesday	Thursday	Friday	Monday	Tuesday	Thursday	Friday
	%	%	%	%	%	%	%	%
Peru, Lima SE	5.05%	5.05%	4.04%	8.71%	2.02%	5.26%	4.04%	1.52%
Singapore, SE of Singapore	7.28%	4.90%	4.48%	6.11%	6.86%	5.07%	4.34%	3.93%
South Africa, Johannesburg SE	6.60%	7.12%	3.96%	5.80%	5.54%	5.54%	3.17%	4.80%
Spain, Barcelona SE	6.06%	5.30%	3.79%	4.34%	4.55%	4.35%	3.79%	2.53%
Sweden, Stockholm SE	7.54%	4.02%	4.02%	10.08%	7.79%	5.23%	3.77%	8.84%
Switzerland, CHE SWX Swiss E.	6.73%	5.20%	4.59%	9.48%	8.26%	5.61%	6.12%	6.42%
Taiwan, Taiwan SE	10.81%	5.11%	4.50%	4.35%	9.61%	10.07%	5.71%	4.24%
Thailand, SE of Thailand	8.95%	4.70%	3.58%	6.79%	5.15%	4.03%	4.25%	5.13%
U.K., SETS	6.44%	5.75%	6.33%	8.27%	5.41%	5.74%	6.10%	6.87%
U.K., LSE-AIM	5.76%	5.84%	5.04%	5.76%	4.96%	5.84%	4.80%	5.12%
U.K., IRS	5.42%	5.42%	8.43%	7.04%	9.64%	6.23%	6.02%	6.63%
U.S., New York SE	8.24%	4.40%	3.96%	7.84%	8.79%	4.83%	4.27%	7.44%
U.S., NASD OTCBB	8.06%	6.05%	5.25%	7.75%	6.47%	4.93%	4.92%	6.47%
U.S., American SE	6.24%	4.43%	4.43%	9.48%	6.84%	5.26%	4.23%	9.66%
U.S., Nasdaq	7.00%	4.33%	5.04%	8.15%	7.29%	5.29%	4.66%	7.92%

References

- Admati AR, Pfleiderer P (1988) A theory of intraday patterns: volume and price variability. *Rev Financ Stud* 1(1):3–40
- Aggarwal R, Rivoli P (1989) Seasonal and day-of-the-week effects in four emerging stock markets. *Financ Rev* 24(4):541–550
- Ajayi RA, Mehdián S, Perry MJ (2004) The day-of-the-week effect in stock returns. *Emerg Markets Finance Trade* 40(4):53–62
- Amihud Y, Mendelson H (1982) Asset price behavior in a dealership market. *Financ Anal J* 38(3):50–59
- Bayar A, Kan OB (2002) Day of the week effects : recent evidence from nineteen stock markets. Central bank review, research and monetary policy department, Central Bank of the Republic of Turkey 2(2) 77–90
- Bekaert G, Harvey CR, Lundblad C (2007) Liquidity and expected returns: lessons from emerging markets. *Rev Financ Stud* 20(6):1783–1831
- Bessembinder H, Hertz MG (1993) Return autocorrelation around nontrading days. *Rev Financ Stud* 6(1):155–190
- Bollerslev T (1986) Generalized autoregressive conditional heteroskedasticity. *J Econ* 31:307–327
- Chang EC, Pinegar JM, Ravichandran R (1993) International evidence on the robustness of the day-of-the-week effect. *J Financ Quant Anal* 28(4):497–513
- Chordia T, Roll R, Subrahmanyam A (2000) Commonality in liquidity. *J Financ Econ* 56(1):3–28
- Chordia T, Roll R, Subrahmanyam A (2001) Market liquidity and trading activity. *J Finance* 56(2):501–530
- Chowdhry B, Nanda V (1991) Multimarket trading and market liquidity. *Rev Financ Stud* 24:483–511
- Condoyanni L, O'Hanlon J, Ward CW (1987) Day of the week effects on stock returns: international evidence. *J Bus* 14(2):159–174
- Connolly RA (1989) An examination of the robustness of the weekend effect. *J Financ Quant Anal* 24(2):133–169
- Cross F (1973) The behavior of stock prices on Fridays and Mondays. *Financ Anal J* 29(6):67–69
- Domowitz I, Glen J, Madhavan A (2001) Liquidity, volatility, and equity trading costs across countries and over time. *International Finance* 221–255

- Dyl EA, Martin SA (1985) Weekend effects on stock returns: a comment. *J Finance* 40(1):347–349
- Eleswarapu VR (1997) Cost of transacting and expected returns in the NASDAQ market. *J Finance* 52:2113–2127
- Engle RF (1982) Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50(4):987–1007
- Fama EF (1965) The behavior of stock-market prices. *J Bus* 38(1):34–105
- Fields MJ (1931) Stock prices: a problem in verification. *J Bus* 4:415–418
- Foster FD, Viswanathan S (1993) Variations in trading volume, return volatility, and trading costs: evidence on recent price formation models. *J Finance* 48:187–211
- French KR (1980) Stock returns and the weekend effect. *J Financ Econ* 8:55–69
- Gibbons MR, Hess P (1981) Day-of-the-week effects and asset returns. *J Bus* 54(4):579–596
- Glosten LR, Jagannathan R, Runkle D (1993) On the relation between the expected value and the volatility of the nominal excess return on stocks. *J Finance* 48:1779–1801
- Hiemstra C, Jones JD (1994) Testing for linear and nonlinear Granger causality in the stock price–volume relation. *J Finance* 49:1639–1664
- Jacoby G, Fowler DJ, Gottesman AA (2000) The capital asset pricing model and the liquidity effect: a theoretical approach. *J Financ Mark* 3:69–81
- Jaffe J, Westerfield R (1985) The week-end effect in common stock returns: the international evidence. *J Finance* 40(2):433–454
- Karpoff JM (1987) The relation between price changes and trading volume: a survey. *J Financ Quant Anal* 22:109–126
- Keim DB, Stambaugh RF (1984) A further investigation of the weekend effect in stock returns. *J Finance* 39(3):819–835
- Lakonishok J, Levi M (1982) Weekend effects on stock returns: a note. *J Finance* 37(3):883–889
- Lakonishok J, Maberly E (1990) The weekend effect: trading patterns of individual and institutional investors. *J Finance* 45(1):231–243
- Lakonishok J, Smidt S (1988) Are seasonal anomalies real? A ninety-year perspective. *Rev Financ Stud* 1(4):403–425
- Lee KH (2005) The world price of liquidity risk. Working paper, Ohio State University.
- Linn S, Lockwood L (1988) Short-term stock price patterns: NYSE, AMEX, OTC. *J Portf Manag* 14(2):30–34
- Merrill A (1966) The behavior of prices on wall street. Analysis Press, Chappaqua
- Ng S, Perron P (2001) LAG length selection and the construction of unit root tests with good size and power. *Econometrica* 69(6):1519–1554
- Pettengill GN (2003) A survey of the Monday effect literature. *Q J Bus Econ* 42(3–4):3–27
- Rogalski RJ (1984) New findings regarding day-of-the-week returns over trading and non-trading periods: a note. *J Finance* 39(5):1603–1614
- Theobald M, Price V (1984) Seasonality estimation in thin markets. *J Finance* 39(2):377–392
- Tong W (2000) International evidence on weekend anomalies. *J Financ Res* 20(9):495–522